FAIR 3D HERITAGE
Documentation, Publication and Preservation of the Digital 3D Heritage on the Web

Explore 3D: Developing a Good Practice Guideline for 3D Digitisation and Open Research Data

Christian Felsner[1], and Alexandre Bovey[2]

[1] ETH Zürich, Switzerland
[2] UNIL (Université de Lausanne), Switzerland

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Abstract

Research objectives

The production of 3D digital objects has become widely accessible in cultural heritage institutions, academic research and education. Producing reliable, reusable and well-documented 3D data, however, remains complex: decisions on capture, processing and documentation strongly influence data quality and usability, and depend on object properties, environmental conditions and intended outcomes. Supporting informed decisions at the start of a project is therefore key to sustainable data management.
"Explore 3D" — a joint project of EPFL, ETH Library Zurich, the Universities of Geneva and Lausanne, and DaSCH in Basel, funded by swissuniversities' Open Science programme — addresses this challenge by developing a lean good practice guideline focused on FAIR and ORD practices for 3D data in the humanities.

Context and related work

3D models are increasingly produced across archives, scientific collections, museums and research projects. Compliance with the FAIR principles is crucial in this context, particularly for the interoperability and reusability of models often developed with considerable effort (European Commission 2022; EUreka3D 2024). The guideline builds on these international frameworks, including the EU's basic principles (European Union 2024) and the Europeana- related EUreka3D guidelines, applying them in a transferable, practice-oriented form.
Recent work highlights the importance of transparent workflows, metadata and paradata for reproducibility and reuse (Manz et al. 2023; Granier et al. 2019; Bajena 2025; van Horik et al. 2025). Existing guidelines, however, address single stages: EUreka3D (2024) formulates general digitisation principles, while PURE3D (Schoueri et al. 2021) and van Horik et al. (2025) focus on publishing finished models. The Explore 3D guideline instead

supports the decisions before capture begins, translating object characteristics and intended outcomes into technology and workflow choices linked to FAIR documentation.

Methods and approach

The guideline structures 3D digitisation workflows around a complexity profile — a structured assessment of the object across four dimensions: object type, size and geometry; location and access; material and optical behaviour; and physical condition and stability. The profile guides technology selection, workflow design and required expertise: reflective or transparent materials, for example, limit optical methods, while immovable objects constrain sensor placement and workflow options. An output-oriented perspective complements this, defining project goals by use, quality requirements and dissemination context.

The framework defines three phases:

1. 3D Capture – selecting acquisition technologies (e.g. photogrammetry, structured light, laser triangulation, LiDAR, CT).
2. Data Processing – optimising geometry, textures (including PBR) and Level of Detail (LOD) models.
3. Publication and Reuse – preparing data for visualisation, research and long-term preservation, with documented metadata and paradata.

Expected results

The resulting guideline includes:

- an overview of capture technologies
- a planning checklist
- recommendations for metadata and paradata
- a workflow from capture to dissemination

The approach is illustrated through a case study: a microscope (E. Leitz GmbH, 1908) from the ETH Library's Collection of Scientific Instruments — a multi-part assembly of reflective brass, a glossy painted base and glass optics — digitised by close-range photogrammetry and published via Sketchfab and Europeana, with metadata and paradata recorded throughout.

Relevance to the conference themes

The guideline addresses documentation practices, metadata and paradata, and sustainable 3D workflows, supporting the creation of reusable 3D datasets for cultural heritage and research.

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